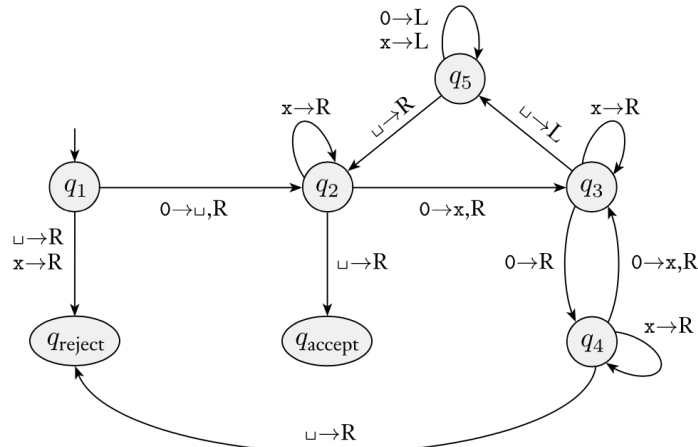


Homework 5

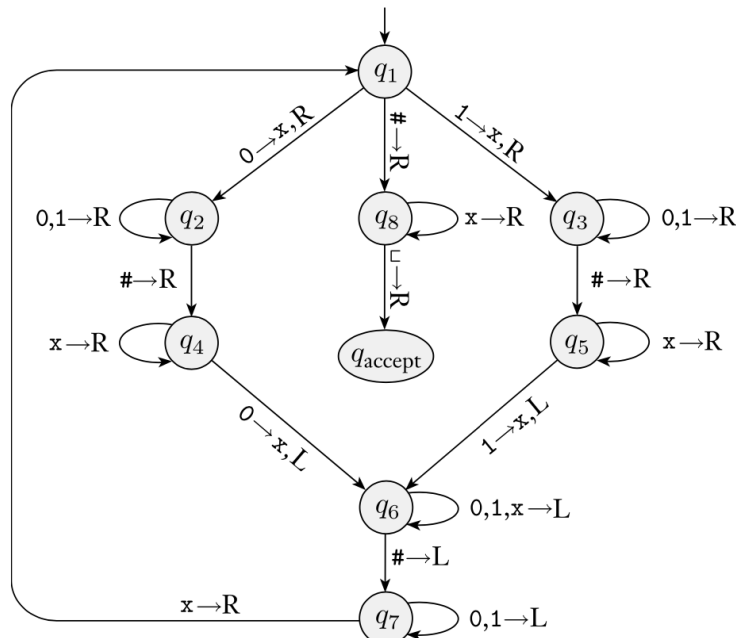
1. Turing machine M_2 decides language $A = \{0^{2^n} \mid n \geq 0\}$, and has the following state diagram:



For the following input strings, give the sequence of configurations M_2 enters.

- a. 0
- b. 00
- c. 000

2. Turing machine M_1 decides language $B = \{w\#w \mid w \in \{0, 1\}^*\}$, and has the following state diagram:



For the following input strings, give the sequence of configurations M_1 enters.

- a. 11
- b. 0#0
- c. 0##0
- d. 1#0